

# JSON

json

**JSON (JavaScript Object Notation)** is a lightweight format used to **store and exchange data** between systems.

- **Human-readable** and **easy to parse**
- **Language-independent**
- **Fast** for data exchange
- Widely used in **APIs, web applications, and microservices**

## Basic structure:

- JSON stores data as **key-value** pairs.
- Data is written inside curly braces {}.
- **Keys** must be **strings** in double quotes.

## JSON supports 6 main data types

### Nested JSON

- **JSON** can contain **objects inside objects** and **arrays inside objects**.
- This structure is very common in API responses.

### Used ?

- **APIs** → Most modern APIs return data in JSON.
- **Configuration files** → Application settings are often stored in JSON.
- **Databases** → Document databases like **MongoDB** store data in **JSON-like format**.
- **Microservices communication** → Services exchange data using JSON.

### Limitations ?

- Does **not** support **comments**
- **No built-in date type**
- **Cannot store functions**
- **Deeply nested JSON** can become **hard to read**.

```
const data = `{
  "name": "Ayushi",
  "age": 26,
  "isStudent": false,
  "address": {
    "street": "123 Main St",
    "city": "Delhi",
    "postalCode": "11001"
  },
  "hobbies": ["travelling", "photography", "cooking"],
  "hasPets": null
};

const dataObject = JSON.parse(data);

console.log(dataObject)
```

### JSON Data Types

” String → "name": "Alex"

# Number → "age": 25

✓ Boolean → "isActive": true

[] Array → "skills": ["Java", "SQL"]

{ } Object → nested JSON inside { }

⊘ Null → "middleName": null

### Java

```
ObjectMapper mapper = new
ObjectMapper();
User user = mapper.readValue
jsonString, User.class);
};
```

### Python

```
import json
data = json.loads(jsonString)
```

